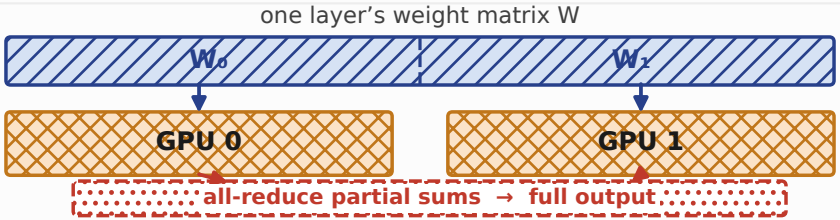
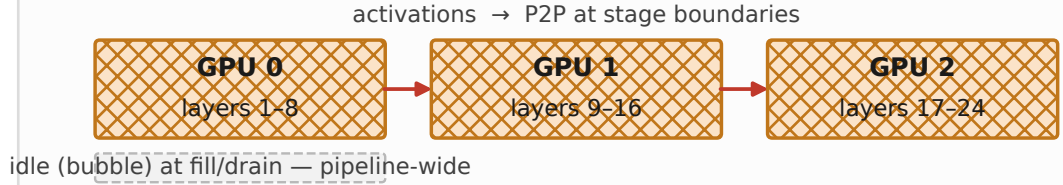
A · Tensor parallel (TP) — splits one layer's weights



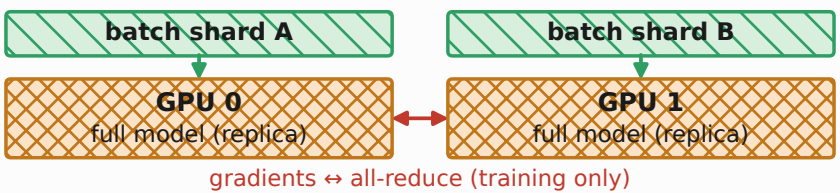
communication: all-reduce every layer → on-node in the canonical config (NVLink)

B · Pipeline parallel (PP) — splits the layer stack



communication-light: only stage boundaries move → spans nodes

C · Data parallel (DP) — splits the data batch



inference: independent replicas, no sync
throughput scales linearly until a shared resource binds

weights (blue) GPU (orange) data (green) comms (red)